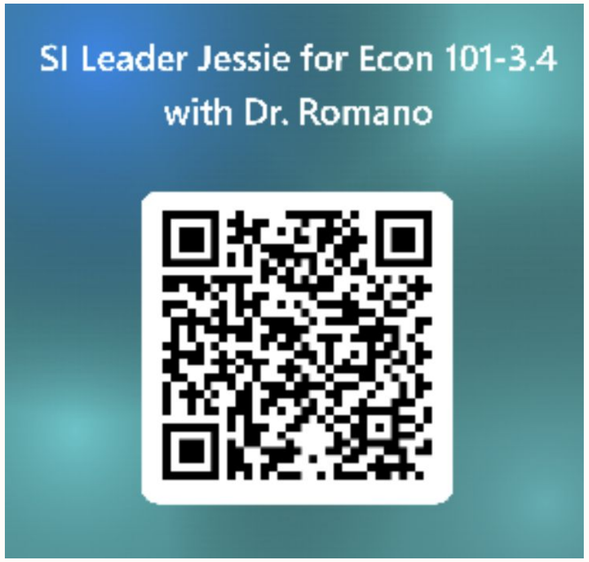
ECON 101
SI Session 11/1





ICE BREAKER: If you were the government, how would you prevent the Tragedy of the Commons?

Externalities

Externality: The uncompensated impact of one's action on the well-being of a bystander

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Allocative Efficiency:
MSC = MSB

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MSC = MSB

Supply: MSC = MPC + MEC
Demand: MSB = MPB +MEB

Assume market demand characterized by P= 42 –QD market supply characterized by P= 6 + 2QS. Further, assume a negative externality equal to $6 per unit.

a) Solve for the equilibrium quantity transacted (Q*) and price transacted (P*) in a market without interventions as well as the efficient quantity Qe.
b) In an appropriately labelled graph, clearly identify the deadweight loss resulting from a market without interventions.
c) Calculate the total deadweight loss
d) Explain how a tax increases total surplus despite (by definition) decreasing both consumer and producer surplus.

D => P = 42 - QD

S => P = 6 + 2QS

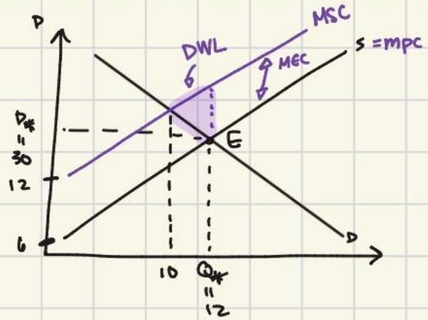
Negative externality = $6.

a) 42 - QD = 6 + 2QS

36 = 3Q

$Q_* = 12$ $P_* = 30$

b)



c) P = 42 - QD P =

42 - QD

30 = 3Qs

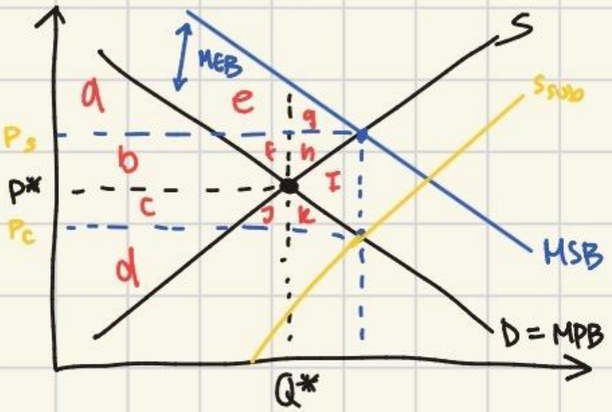
Q = 10

DWL-D (6·2)/2 → 6

d) the tax increases total surplus despite it being a tax because it allows market to consume at what would've been DWL due to the externality. hence it being a corrective tax.

A large city unveils the creation of a new and efficient public transportation system. Experts estimate that the system will impose a positive externality on residents of the city.

a) Draw a diagram of this market, with the externality included.
b) Complete a welfare analysis for the market, imagining a subsidy is put in place equal to the positive externality. Complete the welfare analysis for the market with and without the subsidy.
c) Is there deadweight loss? Why does it arise?



b)

	NO SUB	SUB
CS:	a+b	a+b+j+k
PS:	c+d	b+c+d+f+h
GOV:	—	-(b+f+h+i+j+k+c)
3rd:	e+f	e+f+g+h+i
TS:	a+b+c+d+e+f	a+b+c+d+f+h+e+g

c) Subsidy does not have DWL but without subsidy (H+g) are DWL bc it is what could've been produced.

Tradable Permits

When do we use tradable permits?

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- Negative Externalities
- It creates a New Market

Power Plant	Cost of Reducing 1 ton of pollutant	Current Emissions
North	$200	50 tons
Central	$400	50 tons
South	$600	50 tons

Three power plants (North, Central, and South) each emit 50 tons of pollutants. The EPA sets a cap so that total emissions must fall to 45 tons. Each plant initially receives 35 permits that allow 1 ton of pollution per permit.

a) Determine which plants will buy or sell permits.
b) How many permits will be traded, and what will be the range of prices for a permit sold to each power plant?
c) What is the total cost of meeting the emissions target under trading?
d) What is the total cost of meeting the emissions target without trading?

a) South & Central will buy, North will sell permits

b) North 35p.

Central 35p

South 35p ← wants 15 more p. buys from North at price between $400 - $600



North 20p

Central 35p ← wants 15 more p. buys from North at price between $200 - $400

South 50p



North 5p

central 50p

south 50p.

c) Total cost under trading

→ $200 x 45 = $9000

d) Total cost w/o trading

→ $200 x 15 = $3000

$400 x 15 = $ 6000

$600 x 15 = $ 9,000

= $18000

Coase Theorem:

1)

2)

Coase Theorem:

1) Property rights
2)

Coase Theorem:

1) Property rights
2) Transaction Cost

Types of Goods

What are the 2 Characteristics of Goods?

1)

2)

What are the 2 Characteristics of Goods?

1) Excludability

2)

What are the 2 Characteristics of Goods?

1) Excludability

2) Rivalry in Consumption

Warm Up:

Fill Out the Table: Name +Include examples of each good

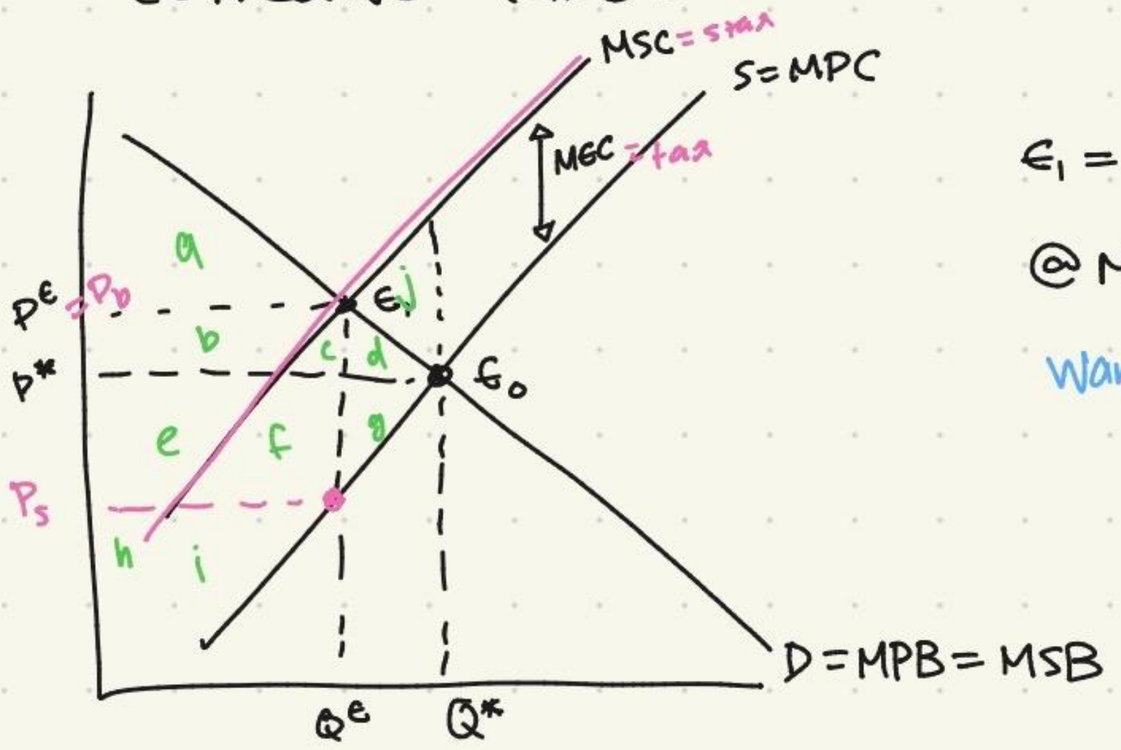
	Excluable	Non-Excludable
Rival		
Not Rival		

Warm Up:

Fill Out the Table: Name +Include examples of each good

	Excluable	Non-Excludable
Rival	Private Goods - Clothes, Food, Cars	Common Resource - Fish in pond - Tragedy of the Commons
Not Rival	Club Goods - Spotify, Netflix	Public Goods -National Defense - Free rider Problem

Corrective Taxes.



Corrective Tax: a tax designed to induce decision makers to respond to _____ social costs (rather than marginal private costs)

Tax = MEC.

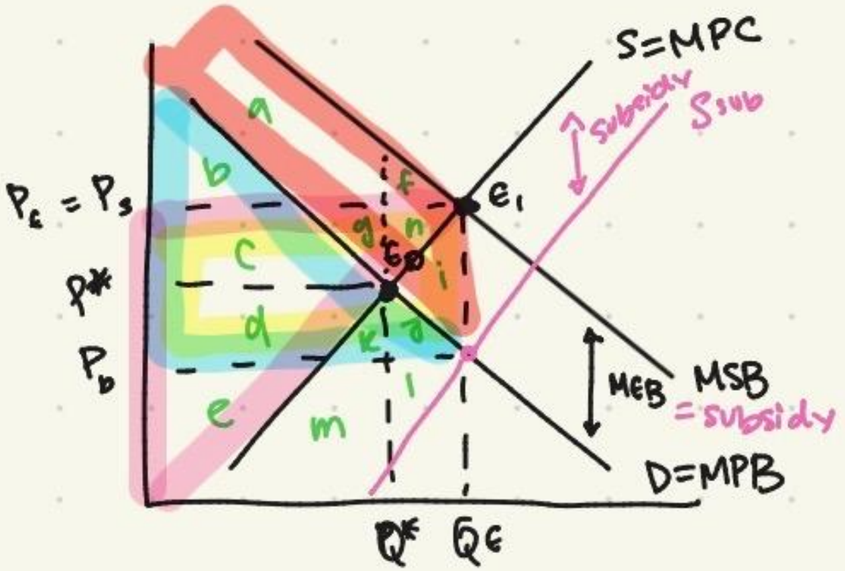
E_1 = Allocative efficency @ MSC = MSB

Want the tax wedge = MEC

TS: A+B+e+h-j | A+B+e+h

	no tax (Q^*, P^*)	tax (Q_E, Q_{tax})	
CS	A+B+C+d	A	
PS	e+h+i+f+g	h+i	bystand: (i+f+C)-
DWL	j | bystand i+f+g+c +J +d		gov: b+C+e+f

Corrective Subsidie



Corrective subsidies: a subsidy designed to induce decision makers to _____ marginal social benefits (rather than private benefits)

	no sub	subsidy
CS	c + b	b + c + d + k + j
PS	d + e	c + d + e + g + h
3rd	a + g	a + g + f + h + i
gov't		c + g + h + l + d + k + j
Total	a + b + c + d + e + g	b + c + d + e + g + h + a + f

b c d ~~k~~ ~~f~~
~~f~~ ~~k~~ e g h
a ~~g~~ ~~f~~ ~~h~~ ~~i~~

Any Questions?

Contact Me!
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